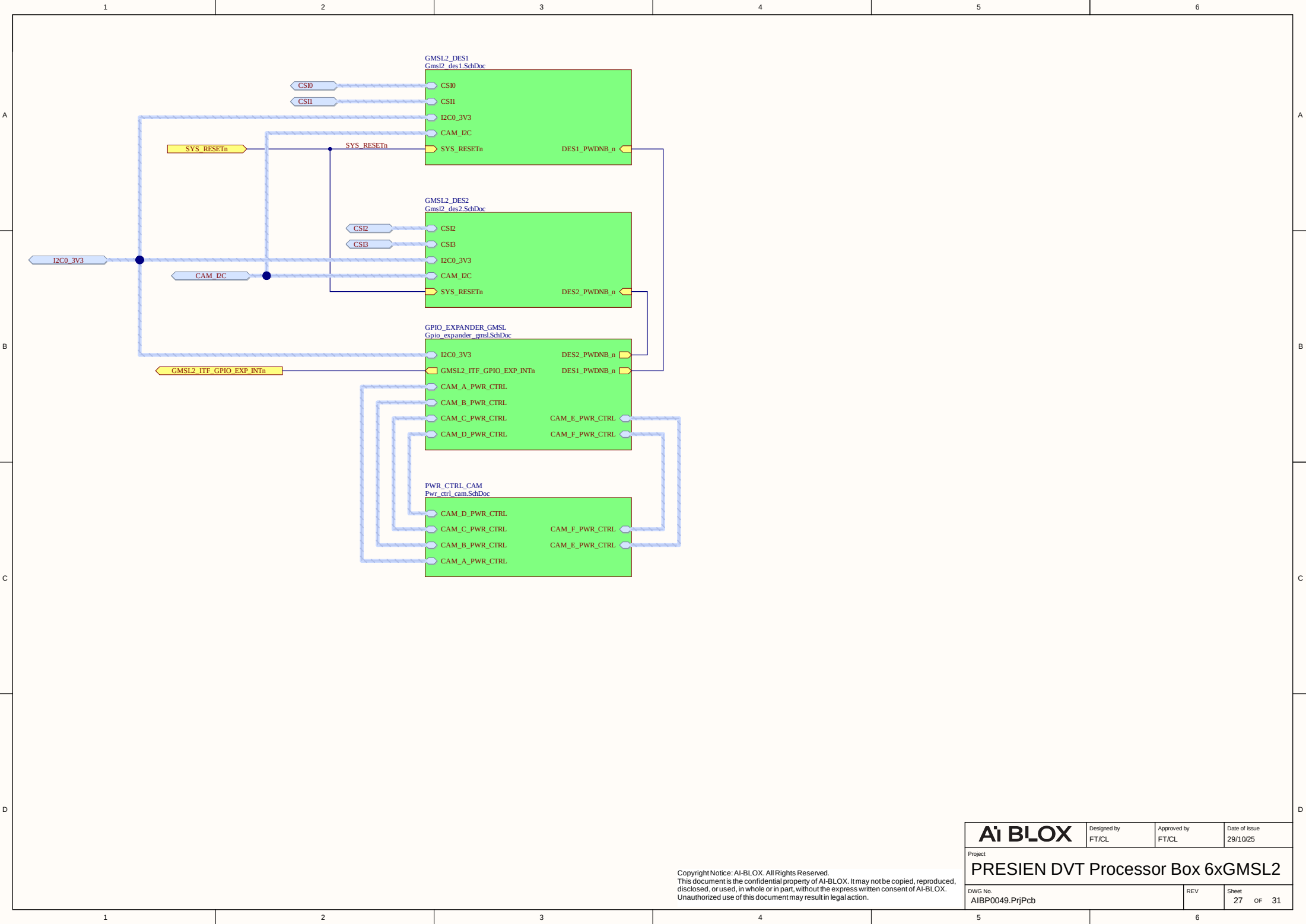
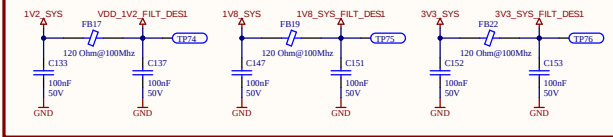




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PWR SUPPLIES FILTERING



Input Data rate calculation (SIO Side):
Camera is ZED X One GS Wide. Output format is RAW10 (from camera dashboard) which means 10bits/pixel.
Preferred capture mode is 960x600@60fps (from Presien Specification)
It leads to 960x600x10x60 = 345.6Mbps per camera video
+ Overhead Protocol

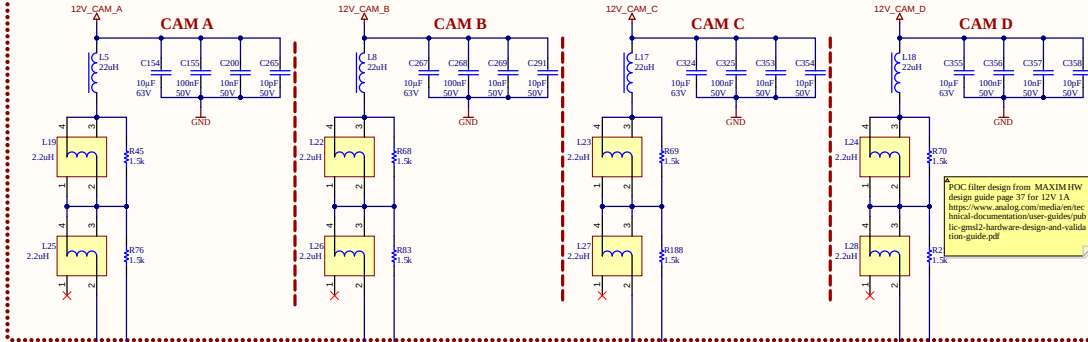
Layout note: CSI DfH Pair should be 100R DfH + All CSI Lane whatever the bus should be length matched all together. See picture below

Table 10-4. MIPI CSI D-PHY Interface Signal Routing Requirements

Parameter	Requirement	Units	Notes
Max Data Rate (per data lane) for High Speed mode	2.5	Gbps	
Max Frequency (for Low Power mode)	10	MHz	
Number of Lanes	1	lane	
Reference plane	GND		
Trace impedance: Diff pair / SE	90-100 / 45-50	Ω	±10%
Via proximity (signal to reference)	< 0.65 (3.0)	mm (in)	
Intra-pair trace spacing	0.15mm	mm	Can be adjusted to meet Differential Impedance.
Trace spacing: Microstrip / Stripline	2x / 2x	dielectric	
Max PCB breakout delay	48	ps	
Max insertion loss		dB	
1 Gbps	3.00		
1.5 Gbps	2.90		
2.5 Gbps	1.92		
Max trace delay / length		ps (mm)	
1 Gbps (Stripline/Microstrip)	2526 (4212) / 2407 (4121)		
1.5 Gbps	2323 (3926) / 2089 (3708)		
2.5 Gbps	900 (1505) / 886 (1505)		
Max intra-pair skew	1	ps	
Max trace delay skew between DQ and CLK	1 / 1.5 / 2.5 Gbps	ps	DQ includes all of the data lines associated with a single clock. This may be 2 differential data lanes for a x2 interface, or 4 differential data lanes for a x4 interface.

Keep critical traces away from other signal traces or unrelated power traces/lines or power supply components

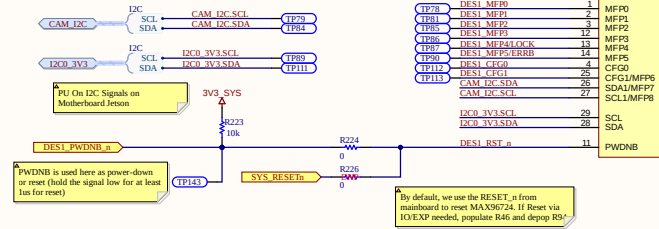
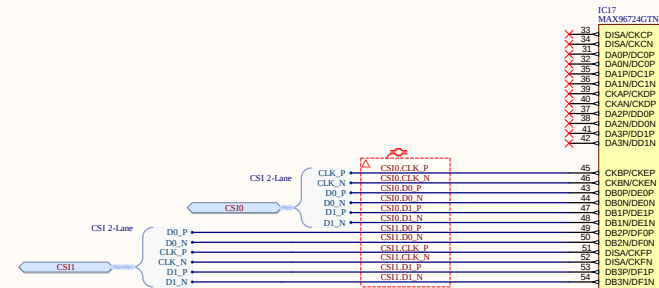
12V INJECTION (PoC)



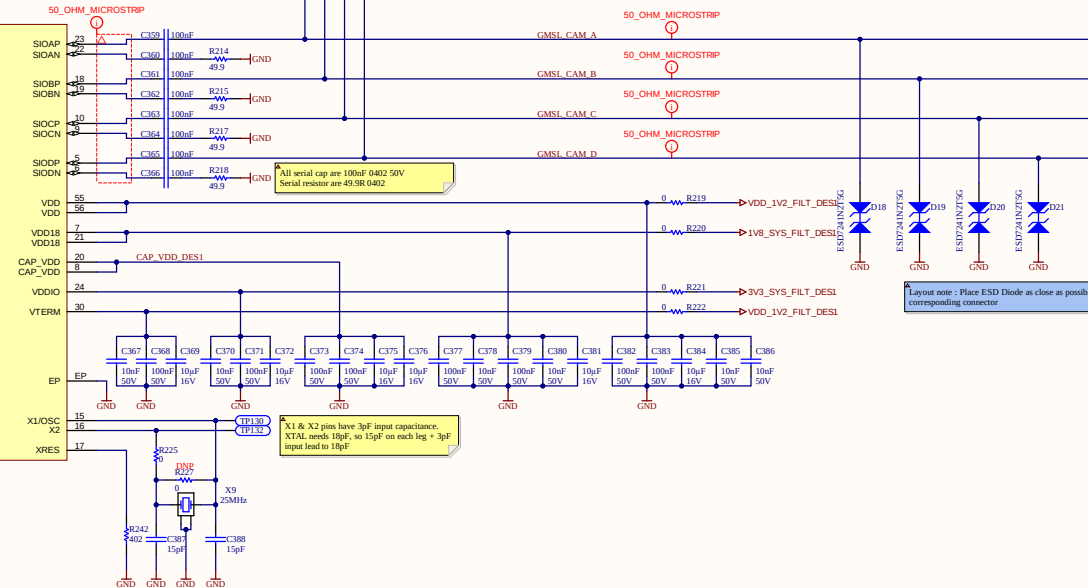
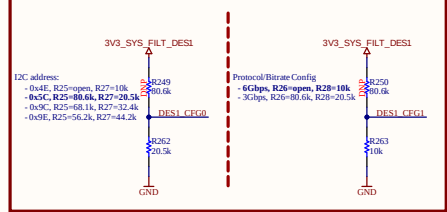
PoC filter design from MAXIM/HW design guide page 37 for 12V 1A
<https://www.analog.com/media/en/technical-documentation/user-guides/pub-lic-gmsl2-hardware-design-and-validation-guide.pdf>

Layout note: Place link capacitor as close as possible to its corresponding SIO Pin

Layout: Remove "copper ring" in internal layer to avoid capacitive effects



CONFIG PINS



All serial cap are 100nF 0402 50V
Serial resistor are 45-3R 0402

X1 & X2 pins have 3pF input capacitance.
XTAL needs 10pF, so 15pF on each leg = 3pF input lead to 10pF

Layout note: Place ESD Diode as close as possible to its corresponding connector

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The schematic diagram shows two identical power supply sections for CAM E and CAM F. Each section starts with a 12V input labeled 12V_CAM_E or 12V_CAM_F. This input goes through a series of capacitors: L29 (2.2uHf), C395 (100nF 50V), C396 (100nF 50V), C397 (10uF 50V), and C398 (10uF 50V). The output of this filter network is connected to a transformer (L31 for CAM E, L30 for CAM F) with a 2.2uHf primary winding. The secondary windings are connected to a bridge rectifier (R367 for CAM E, R368 for CAM F) followed by another filter capacitor (C400, 100nF 50V). The final output is connected to a second transformer (L33 for CAM E, L34 for CAM F) with a 2.2uHf primary winding. The secondary windings are connected to a bridge rectifier (R399 for CAM E, R300 for CAM F) followed by a final filter capacitor (C401, 100nF 50V). The ground connection is labeled GND.

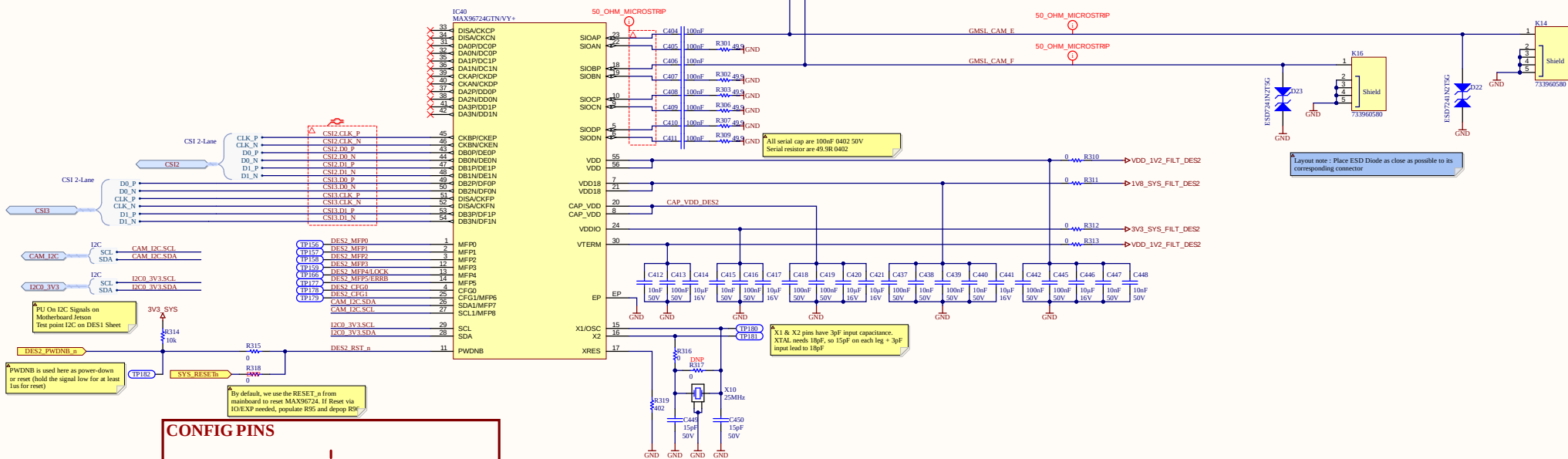
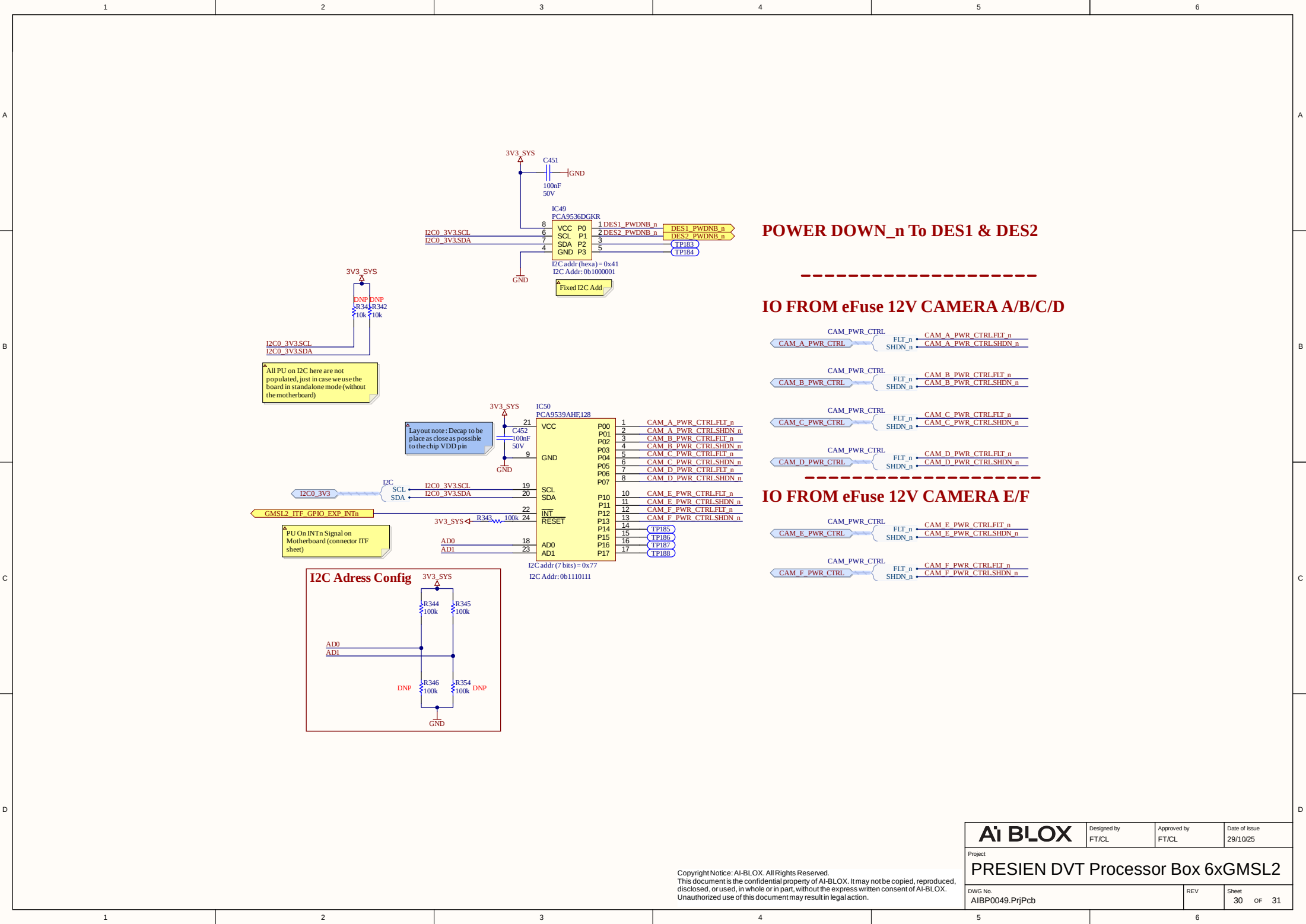
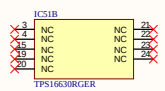


Figure 10 shows the schematic diagram of the 3V3 SYS_FILT_DES2 filter circuit. The circuit is divided into two sections by a dashed red line. The left section, labeled "I2C address:", shows the 3V3 SYS_FILT_DES2 pin connected to a 10k resistor (R3320) and a 10k resistor (R3339) to GND. The right section, labeled "Protocol/BiRate Config", shows the 3V3 SYS_FILT_DES2 pin connected to a 10k resistor (R3321) and a 10k resistor (R340) to GND. The circuit is powered by 3V3 and GND.

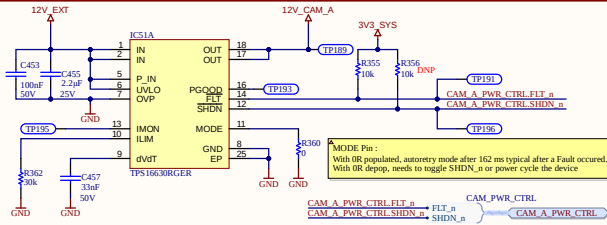
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CAMERA A eFUSE 12V



Item is set by external resistor to GND with this formula :
 $I_{lim} = 18R_{gnd}$ with I_{lim} in Amps, R_{gnd} in kOhm.
 500mA protection leads to $R_{gnd} = 18/0.5 = 36k\Omega$
 Slew Rate defined by
 $T_{dvdt} = 1000 \times 20.8 \times 10^{-9} \times C_{dvdt} \rightarrow$ With 33nF, leads to 8.2ms



Power Consumption of Camera ZED X ONE GS WIDE is :

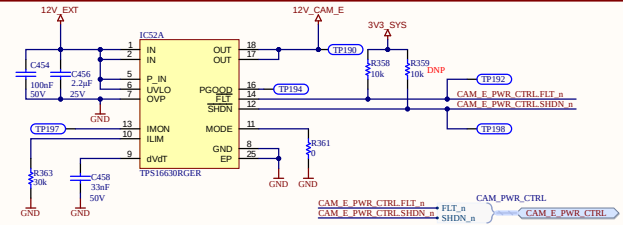
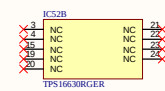
- Min : 0.7W <==> 59mA @12V
- Max : 0.94W <==> 79mA @12V

For reference :

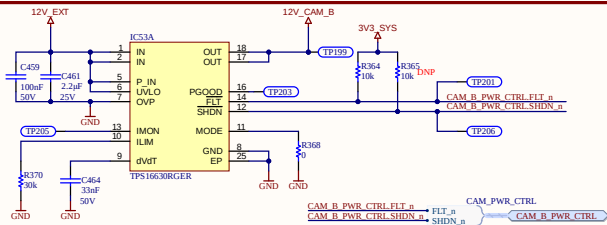
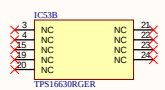
https://www.stereolabs.com/docs/get-started-with-zed-link/power_requirements

For later design change, if we need to reduce the price, use the following eFuse : TPS1641x

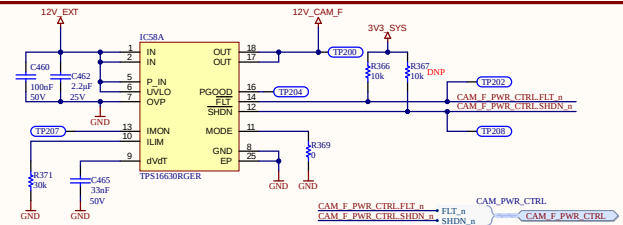
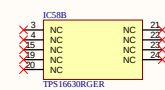
CAMERA E eFUSE 12V



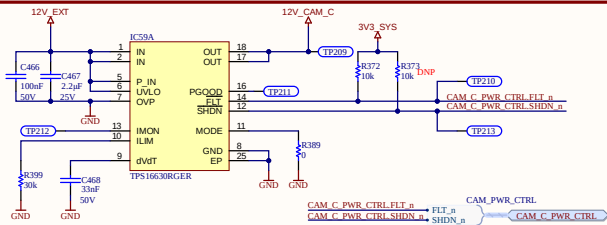
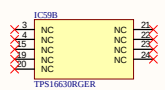
CAMERA B eFUSE 12V



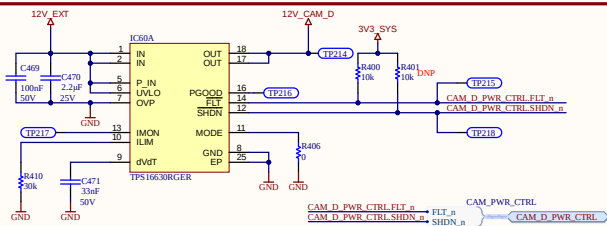
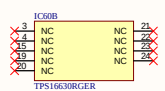
CAMERA F eFUSE 12V



CAMERA C eFUSE 12V

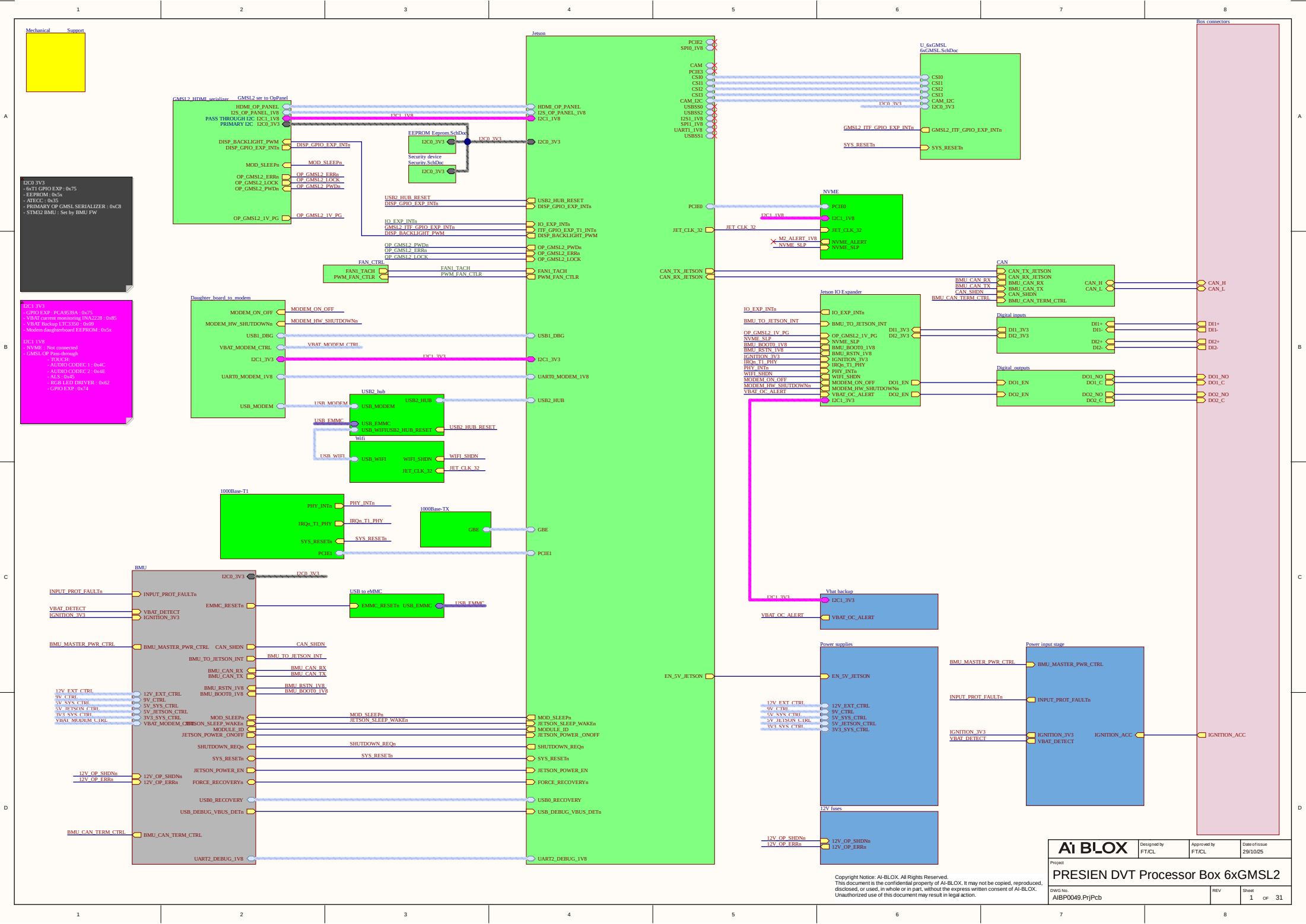


CAMERA D eFUSE 12V



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